

Server Fleet Runbook — Game Title

Fictional demo document for Studio Agent Nexus. Synthetic content, safe for public demonstration. Operational runbook for the dedicated game-server fleet that hosts Game Title matches at Game Studio.

1. Purpose

This runbook covers how the dedicated game-server fleet is operated: regions, allocation, autoscaling, deployment, rollback, and on-call response. It is the operational companion to the Backend Architecture Overview.

2. Fleet topology

Game Title runs dedicated server instances grouped into regional pools. Each match is hosted on exactly one instance, allocated on demand when matchmaking forms a match. Instances are stateless beyond the lifetime of the match they host.

- Regions: NA-East, NA-West, EU-Central, and AP-Southeast.
- Each region maintains a warm pool of pre-booted instances to absorb demand spikes without cold-start latency.
- An allocator service hands a ready instance to matchmaking on request and marks it busy for the match duration.

3. Capacity and sizing

Per-instance capacity is derived from the simulation cost. For sizing, the fleet model assumes each server instance runs the simulation at 60 Hz and budgets CPU headroom accordingly. Instance counts per region are scaled from expected concurrent matches at peak.

Note: capacity numbers must be reconciled with the authoritative tick rate stated in the Backend Architecture Overview before the load test.

4. Autoscaling

The fleet autoscales on warm-pool depletion and queued-allocation pressure. When the warm pool falls below a regional threshold, the allocator requests additional instances; idle instances above the threshold are drained after a cooldown.

- Scale-up trigger: warm pool below regional minimum for 60 seconds.
- Scale-down trigger: warm pool above regional maximum for 5 minutes.
- Scale-up step and cooldown are tuned per region.

Open gap: the maximum fleet size is not defined. There is currently no hard upper bound on how many instances autoscaling may request, so capacity planning, cost ceilings, and saturation alerts cannot be set. This must be resolved before the beta load test.

5. Deployment

Server builds are promoted dev → staging → prod through the CI/CD pipeline. Deploys are rolling: new instances boot the new build into the warm pool while in-flight matches drain on the old build. No live match is interrupted by a deploy.

6. Rollback

If a bad build is detected, the allocator is pinned to the last-known-good build and new instances boot from it. Instances on the bad build are drained as their matches end. Rollback does not evict live matches.

7. On-call response

- Capacity alert (warm pool exhausted): verify autoscaling is firing; manually raise regional minimums if demand exceeds the model.
- Allocation failures: check allocator health and regional instance supply.
- Bad build: trigger rollback and notify the incident commander.

Primary on-call for the fleet is the Infra owner listed in the Team Directory. Severity calls and external communication are owned by the incident commander.